

Lemma.

Let K/F be a field extension, and let $\alpha \in K$ be algebraic over F . Then, for any $\sigma \in \text{Aut}(K/F)$, $\sigma\alpha$ is a root of the minimal polynomial for α over F .

Proof. Let $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0 \in F[x]$ be the minimal polynomial for α . Then, $f(\alpha) = 0$, so $\sigma(f(\alpha)) = 0$. (The field homomorphism preserves zero).

Now, $0 = \sigma(f(\alpha)) = \sigma(\alpha^n + a_{n-1}\alpha^{n-1} + \dots + a_1\alpha + a_0)$

$$\begin{aligned} \sigma \text{ fixes } F \quad \hookrightarrow &= (\sigma\alpha)^n + \sigma(a_{n-1}) \cdot (\sigma\alpha)^{n-1} + \dots + \sigma(a_1) \cdot (\sigma\alpha) + \sigma(a_0) \\ &= (\sigma\alpha)^n + a_{n-1}(\sigma\alpha)^{n-1} + \dots + a_1(\sigma\alpha) + a_0 \\ &= f(\sigma\alpha). \end{aligned}$$

Prop. Let $H \leq \text{Aut}(K)$ be a subgroup of the group of all automorphisms of K .

Then, the collection F of elements of K fixed by all the elements of H is a subfield.

Proof. First, given $\sigma \in \text{Aut}(K)$, the set $F_\sigma = \{a \in K \mid \sigma(a) = a\}$ is subfield, because:

Let $a, b \in F_\sigma$ be given. Then, $\sigma(a-b) = \sigma(a) - \sigma(b) = a-b$, and if $b \neq 0$, then $\sigma(a \cdot b^{-1}) = \sigma(a) \cdot \sigma(b)^{-1} = a \cdot b^{-1}$, so the claim is proved.

And now, since the arbitrary intersection of fields is field, so proved. ~~QED~~
Such a subfield is called the fixed field of $S \subseteq \text{Aut}(K)$.

Prop. Let K be a field. Then,

If $F_1 \leq F_2 \leq K$, then $\text{Aut}(K/F_2) \leq \text{Aut}(K/F_1)$.

If $H_1 \leq H_2 \leq \text{Aut}(K)$, then $F_{H_2} \leq F_{H_1}$.

In other say, The inclusion relation of automorphisms and fixed field is reverse.

Lemma.

Def. A finite extension K/F is called a Galois Extension if:

$$|Aut(K/F)| = [K:F].$$

In such a case, denote $Gal(K/F) = Aut(K/F)$.

Def. A character χ of a group G with values in a field L is a group homomorphism

$$\chi: G \rightarrow L^\times$$

The characters $\chi_1, \dots, \chi_n$ of G are said to be linearly independent over L if:

$$\alpha_1 \chi_1 + \dots + \alpha_n \chi_n \equiv 0 \text{ for some } \alpha_i \in L, \text{ then } \alpha_1 = \dots = \alpha_n = 0.$$

[Thm. [Dedekind-] Independence of Characters Theorem]

If $\chi_1, \dots, \chi_n: G \rightarrow L^\times$ are distinct characters of G with values in L , then they are linearly independent over L .

Proof. Suppose that the characters are linearly dependent.

That is, there exists non-trivial representation: for some $\alpha_1, \dots, \alpha_n \in L$,

$$\alpha_1 \chi_1 + \dots + \alpha_n \chi_n \equiv 0 \text{ where } \alpha_1, \dots, \alpha_n \text{ are not all zero.}$$

So, the set $\{m \in \mathbb{N} \mid \text{for some } b_1, \dots, b_m \in L \setminus \{0\}, \text{ and } \sigma \in S_n, b_1 \chi_{\sigma(1)} + \dots + b_m \chi_{\sigma(m)} \equiv 0\}$ is non-empty, thus it has a minimal element by the well-ordering principle.

$m \in \mathbb{N}$ with $b_1 \chi_1 + \dots + b_m \chi_m \equiv 0$, the indices are renumbered, $b_i \in L \setminus \{0\}$.

Since the characters are distinct, $\chi_1(g_0) \neq \chi_m(g_0)$, for some $g_0 \in G$.

Given $g \in G$, observe that

$$0 = b_1 \chi_1(g_0 g) + \dots + b_m \chi_m(g_0 g) = b_1 \chi_1(g_0) \chi_1(g) + \dots + b_m \chi_m(g_0) \chi_m(g)$$

Meanwhile,

$$0 = \chi_m(g_0) \cdot 0 = \chi_m(g_0) \cdot (b_1 \chi_1(g) + \dots + b_m \chi_m(g)) = b_1 \chi_m(g_0) \chi_1(g) + \dots + b_m \chi_m(g_0) \chi_m(g),$$

Subtracting gives $[\chi_m(g_0) - \chi_1(g_0)] \cdot b_1 \chi_1(g) + \dots + [\chi_m(g_0) - \chi_{m-1}(g_0)] \cdot b_{m-1} \chi_{m-1}(g) = 0$, because the last term is terminated. But, since $[\chi_m(g_0) - \chi_1(g_0)] \neq 0$, it is another representation, but that is contradiction to the minimality of m .

[Corollary. Distinct Field Automorphisms are linearly independent.]

[Thm.

Let $G = \{\sigma_1 (= \text{id}), \sigma_2, \dots, \sigma_n\}$ be a subgroup of $\text{Aut}(K)$, and let $F = K^G$.

Then, $[K:F] = |G| = n$.

Proof. To show the equality, we will show that neither $|G| > [K:F]$ nor $|G| < [K:F]$. Suppose that $n > [K:F]$ and let $w_1, \dots, w_m$ be a basis for K over F . (So, $m = [K:F]$)

Consider a system

$$\begin{aligned} \sigma_1(w_1)x_1 + \sigma_2(w_1)x_2 + \dots + \sigma_n(w_1)x_n &= 0 \\ \vdots & \\ \sigma_1(w_m)x_1 + \sigma_2(w_m)x_2 + \dots + \sigma_n(w_m)x_n &= 0 \end{aligned} \quad \left(\text{or } \left(\sigma_j(w_i) \right)_{i,j} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = 0 \right)$$

By the assumption, $n > m$, so the system has a non-trivial solution $\beta_1, \dots, \beta_n$ in K .

(By the dimension theorem, nullity $A = n - \text{rank } A \geq n - m > 0$.)

Given $a_1, \dots, a_m \in F$, these are fixed by $\sigma_i \in G$, by the definition of F .

Multiplying a_i to the i th row of the equation, then

$$\sigma_1(a_1 w_1) \beta_1 + \dots + \sigma_n(a_1 w_1) \beta_n = 0$$

$$\sigma_1(a_m w_m) \beta_1 + \dots + \sigma_n(a_m w_m) \beta_n = 0.$$

So, adding all the equations i for $\alpha = a_1 w_1 + \dots + a_m w_m$,

$$\beta_1 \sigma_1(\alpha) + \dots + \beta_n \sigma_n(\alpha) = 0.$$

Since $a_1, \dots, a_m$ was arbitrary and $w_1, \dots, w_m$ forms basis for K over F , the α is arbitrary element of K . Moreover, since $\beta_1, \dots, \beta_n$ are not all zero, it contradicts the G is linearly independent, by the Dedekind Character Theorem. Now, suppose that $n < [K:F]$. That is, there are more than n F -linearly independent elements, say $d_1, \dots, d_{n+1}$. Consider a system of $n+1$ unknowns,

$$\sigma_1(d_1)x_1 + \dots + \sigma_n(d_{n+1})x_{n+1} = 0$$

$$\sigma_n(d_1)x_1 + \dots + \sigma_n(d_{n+1})x_{n+1} = 0. \text{ Then, it has } n \text{ equation and } n+1 \text{ unknowns}$$

So there is non-trivial solution $\beta_1, \dots, \beta_{n+1}$ in K .

Moreover, since $\sigma_1 = \text{id}$ and $d_1, \dots, d_{n+1}$ are F -linearly independent, not all β_i lies in F .

Corollary. Let K/F be any finite extension. Then,

$$|\text{Aut}(K/F)| \leq [K:F]$$

In Particular, the equality holds if and only if $F = K^{\text{Aut}(K/F)}$
if and only if K/F is Galois extension
if and only if K is splitting field of separable polynomial over F .

Proof.

Corollary.

Let G be a finite subgroup of $\text{Aut}(K)$. Then

$$\text{Aut}(K/K^G) = G.$$

And, if $G_1 \neq G_2$, then $K^{G_1} \neq K^{G_2}$.

Fundamental Theorem of Galois Theory.

Let K/F be a Galois Extension. Denote $G = \text{Gal}(K/F)$.

1). A map

$$\mathcal{Q}: \{H \leq \text{Gal}(K/F) \mid H \text{ is subgroup}\} \rightarrow \{F \leq E \leq K \mid E \text{ is intermediate field}\}$$

$$: H \mapsto K^H$$

is bijection, with the inverse $\mathcal{Q}^{-1}(E) = \text{Gal}(K/E)$.

2). Given $H_1, H_2 \leq G$,

$$H_1 \leq H_2 \text{ if and only if } K^{H_2} \leq K^{H_1}.$$

3). Given $H \leq G$,

$$[K:K^H] = |H| \quad \text{and} \quad [K^H:F] = |\text{Gal}(K/F):H|$$

$$\begin{array}{c} K \\ / \quad | \quad [H] \\ |G| \quad K^H \\ \backslash \quad | \quad [G:H] \\ F \end{array}$$

4). Given $H \leq G$, K/K^H is always Galois, with $\text{Gal}(K/K^H) = H$.

Moreover, K^H/F is Galois if and only if $H \leq G$ is normal.

If K^H/F is Galois, then $\text{Gal}(K^H/F) \cong \text{Gal}(K/F)/H$.

Further, there exists one-to-one correspondence between

$$\{\varphi: K^H \rightarrow \overline{K} \mid \varphi \text{ is embedding which fix } F\} \leftrightarrow \{\sigma H \mid \sigma \in G\},$$

5). Given $H_1, H_2 \leq G$, then

$$K^{\langle H_1, H_2 \rangle} = K^{H_1} \cap K^{H_2} \quad \text{and} \quad K^{H_1 H_2} = K^{H_1} K^{H_2}$$

$$G = \text{Gal}(K/F)$$

$$\begin{array}{c} G_1 \\ \vdots \\ G_{(2,1)} \quad G_{(2,2)} \end{array}$$

$$\leftrightarrow$$

$$\begin{array}{c} F = K^G \\ \swarrow \quad \searrow \\ K^{G_1} \quad K^{G_2} \\ \vdots \quad \vdots \\ K^{G_{(2,1)}} \quad K^{G_{(2,2)}} \end{array}$$

$$\{e\}$$

$$K = K^{\{e\}}$$

